

Wheat Disease Classification Model

Presented By:

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Introduction

Problem:

- Early detection of disease in crops without automation is tedious
- Existing automated solutions require specialized, inaccessible technology

Wheat plant disease detection impacts:

- Farmers' profits
- Global food production

Agenda

Design Overview
The process to our model

**Model Training and
Tuning**
Maximizing functionality of
the model



Background
The story behind the
problem

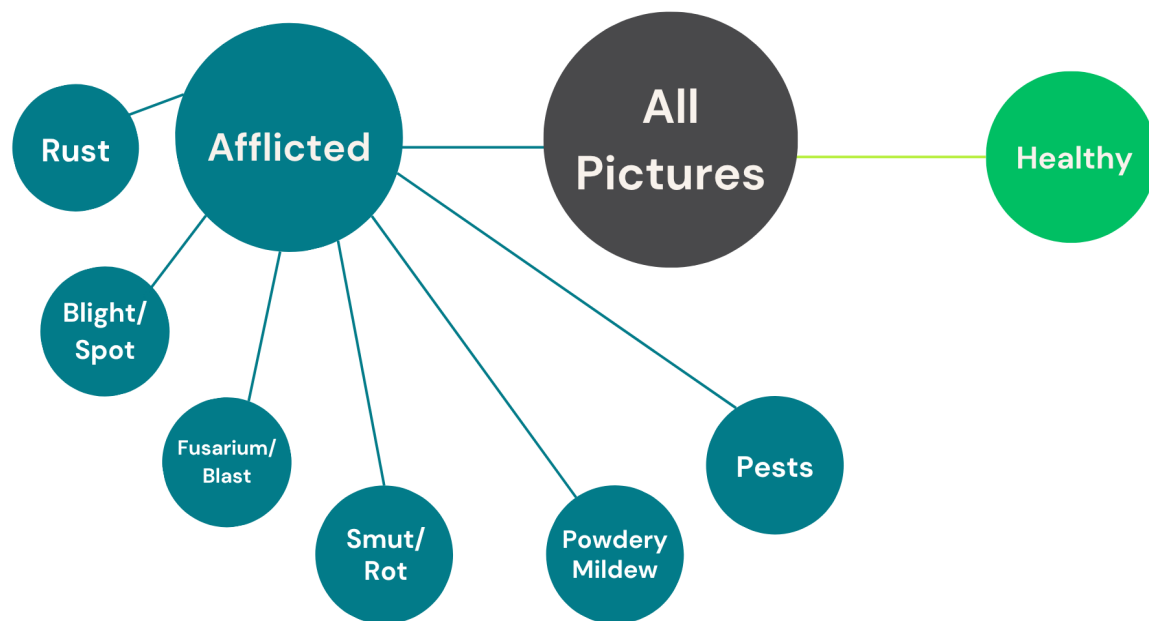
Data Exploration
Initial look at the data



Background:

Using Deep Learning for Image-Based Plant Disease Detection. (Mohanty et al.)	Cassava Disease Detection Using Convolutional Neural Networks. (Sambasivam & Opiyo)	Enhanced multiscale plant disease detection with the PYOLO model innovations. (Zhang et al.)
99.35% accuracy	94.32% accuracy	~98% accuracy
All images were produced in an incredibly sterile fashion	Study only done on Cassava (not wheat)	Requires hyperspectral imaging

Design Overview



Key Components:

- Sorting
- Balancing
- Preprocessing
- Training
- Fine-tuning

Preprocessing and Exploratory Data Analysis

Data pre-processing was done using skimage

- Converting to arrays
- Flattening
- Compiling into a large DataFrame for training

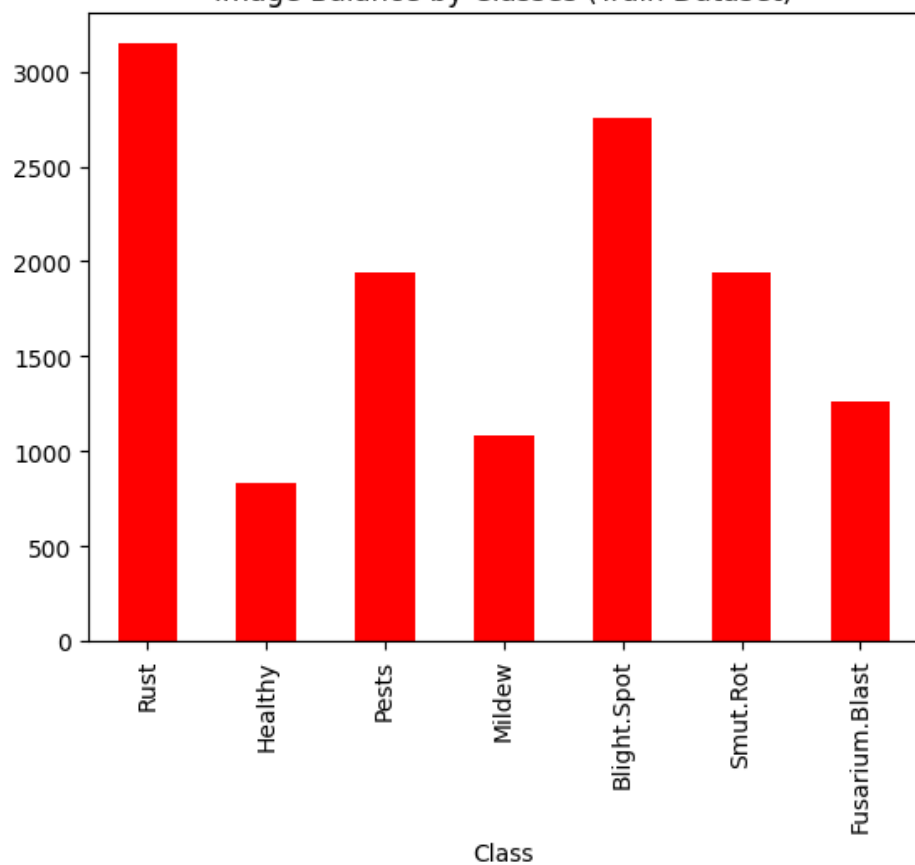
EDA looked at various components of the data

- Balance of categories
- Color by category
- Overall importance of color

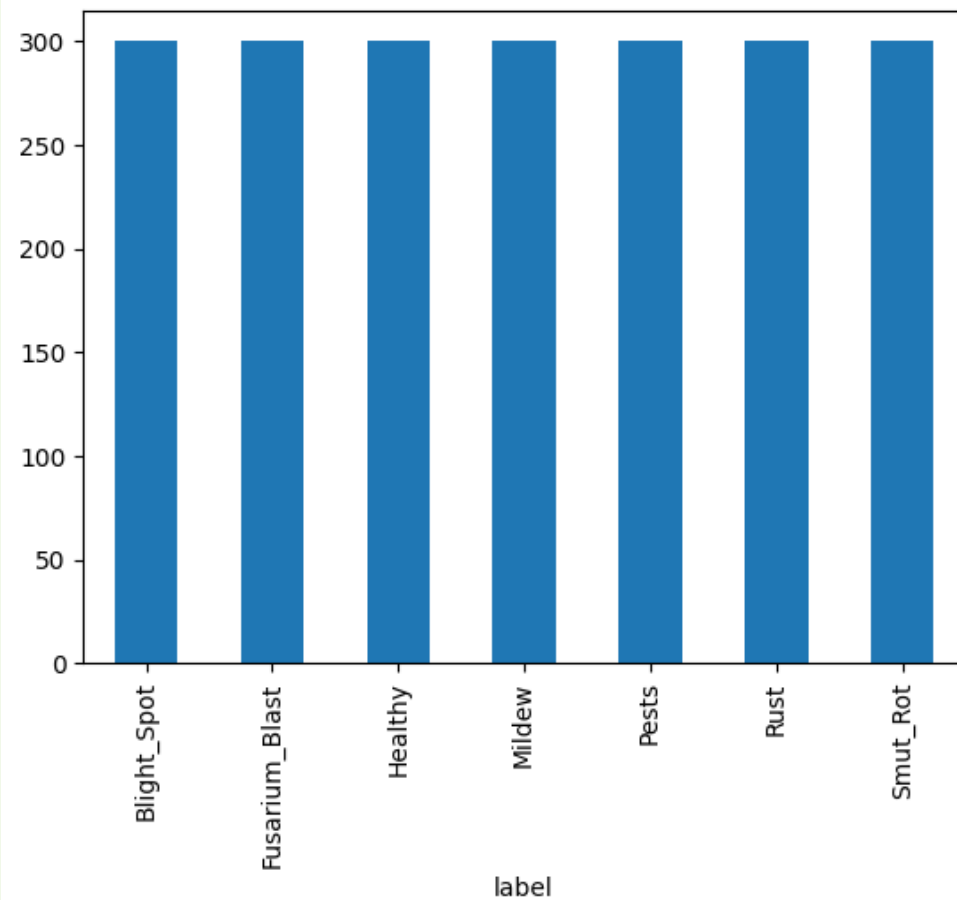
Balance of Categories

Using existing training split:

Image Balance by Classes (Train Dataset)



Resampled data:



Color By Category

Image Chunk 1-100



Image Chunk 101-200



Image Chunk 201-300



Image Chunk 301-400



Image Chunk 401-500



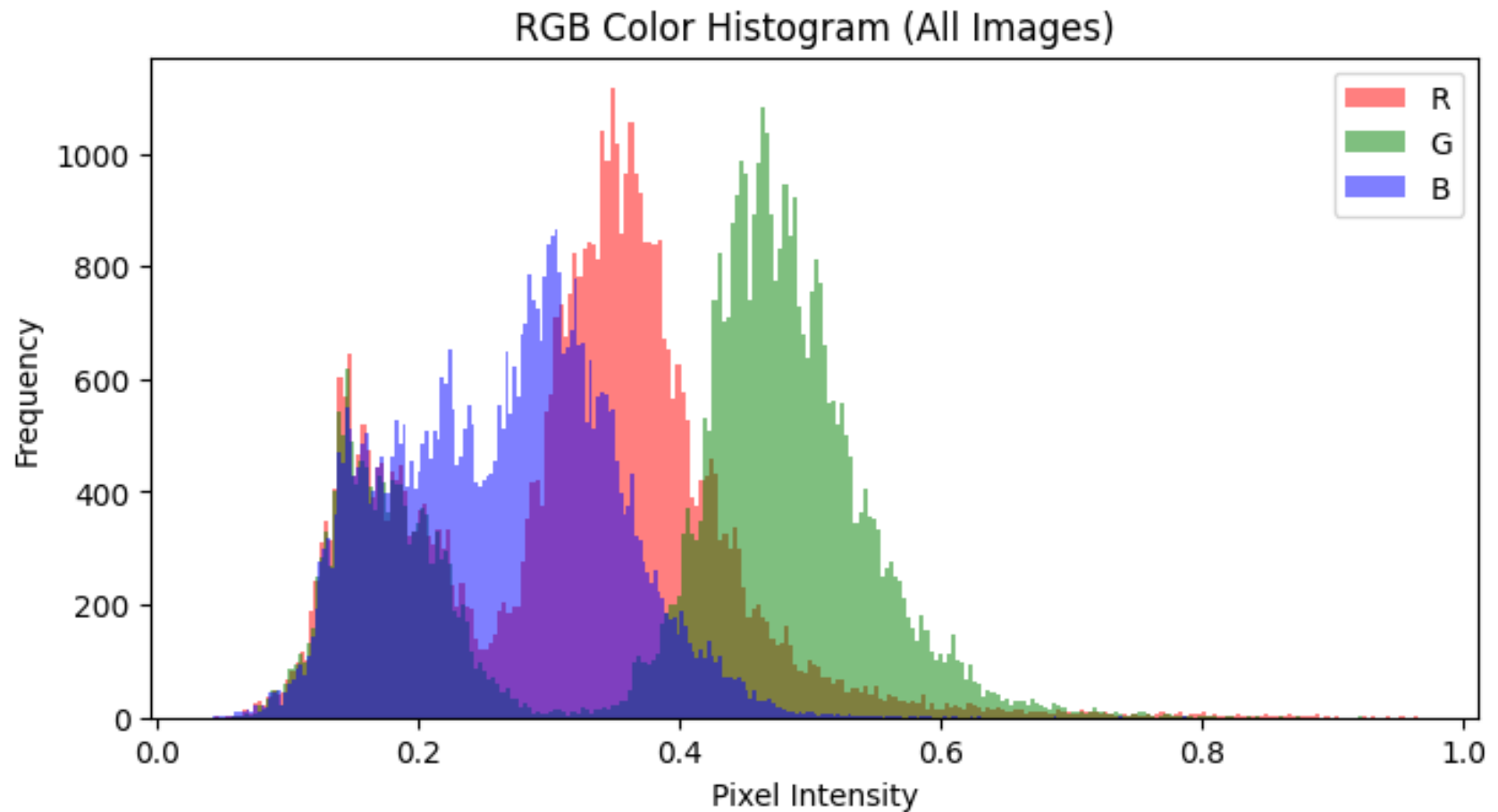
Image Chunk 501-600

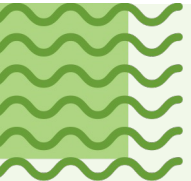


Image Chunk 601-700



Histogram of Colors





Training/Modeling

Data pre-processing was done using skimage

- Converting to arrays
- Flattening
- Compiling into a large DataFrame for training

Training was done with a variety of models, and many were fine-tuned

- Logistic regression
 - KNN
 - SVM: tuned with GridSearchCV
 - CNN: tuned through increased training
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